

Lemma 3. For all $x \in \Sigma^*$, $\hat{S}'([P], x) = [\hat{S}(P, x)]$

Lemma 2. $p \in F$ iff $[p] \in F'$.

Theorem. $L(M/\approx) = L(M)$.

Proof. For $x \in \Sigma^*$,

$x \in L(M/\approx)$ iff $\hat{S}'(\delta', x) \in F'$

iff $\hat{S}'([\delta], x) \in F'$ [Defn. of δ']

iff $[\hat{S}(\delta, x)] \in F'$ [Lemma 3]

iff $\hat{S}(\delta, x) \in F$ [Lemma 2]

iff $x \in L(M)$ [Defn. of acceptance]

What if you do the quotient construction again on $M/\approx$?

$$[p] \sim [q] \text{ iff } \forall x \in \Sigma^* (\hat{s}'([p], x) \in F' \text{ iff } \hat{s}'([q], x) \in F')$$

Use $\sim$ to denote the equivalence relation on Q' to distinguish it from the relation $\approx$ on Q .

$$[p] \sim [q] \Rightarrow \forall x (\hat{s}'([p], x) \in F' \text{ iff } \hat{s}'([q], x) \in F') \quad [\text{Defn. of } \sim]$$

$$\Rightarrow \forall x ([\hat{s}(p, x)] \in F' \text{ iff } [\hat{s}(q, x)] \in F') \quad [\text{Lemma 3}]$$

$$\Rightarrow \forall x (\hat{s}(p, x) \in F \text{ iff } \hat{s}(q, x) \in F) \quad [\text{Lemma 2}]$$

$$\Rightarrow p \approx q \Rightarrow [p] = [q]$$

$\therefore$ Two equivalent states of $M/\approx$ are equal

and $\sim \subseteq Q' \times Q'$ is the identity relation.

Algorithm for DFA minimization

Algorithm takes as input a DFA $M = (Q, \Sigma, \delta, s, F)$ and marks pairs of states (unordered)

A pair $\{p, q\}$ will be marked when the procedure detects that p and q are not equivalent.

Algorithm.

1. All pairs of states $\{p, q\}$ are initially unmarked.
2. if $p \in F$ and $q \notin F$ or vice versa, mark $\{p, q\}$.
- 3 Repeat until no change in marking
- 4 if $\exists \{p, q\}$ -unmarked such that
 $\{\delta(p, a), \delta(q, a)\}$ is marked for some $a \in \Sigma$
 then mark $\{p, q\}$.
- 5 $p \approx q$ iff $\{p, q\}$ is not marked.

$$p \approx q \text{ iff } \forall x \in \Sigma^* (\hat{\delta}(p, x) \in F \text{ iff } \hat{\delta}(q, x) \in F)$$

Algorithm.

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- 3 Repeat until no change in marking
- 4 if $\exists \{p, q\}$ -unmarked such that
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Termination. There are atmost $\binom{n}{2}$ pairs.

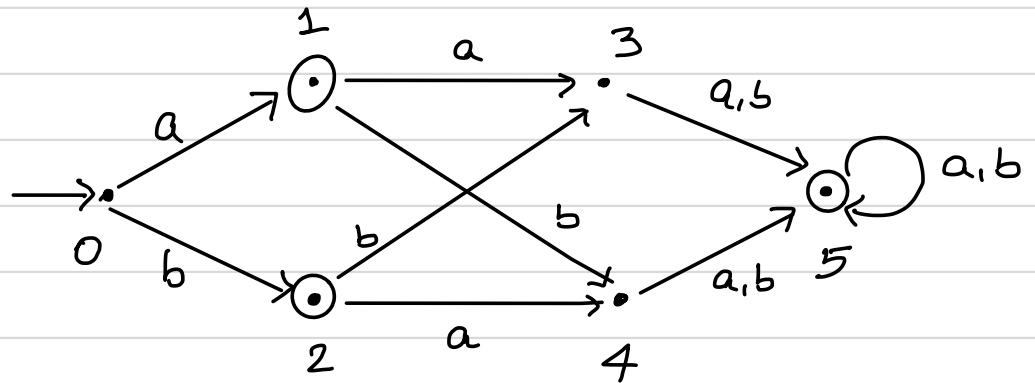
Step 4 marks at least one pair in each iteration

$\therefore$ Repeat step [line 3] can have atmost $\binom{n}{2}$ iterations.

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Example.

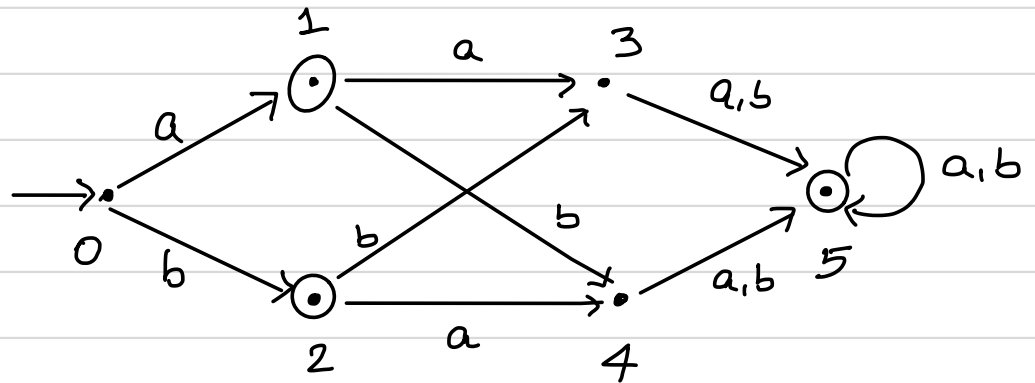


0	1	2	3	4	5
✓					
✓					
	✓	✓			
	✓	✓			
✓				✓	✓

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Example.



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1

✓

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2

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3

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✓

✓

$\{1, 5\} \xrightarrow{a} \{3, 5\}$.

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4

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✓

✓

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$\{2, 5\} \xrightarrow{a} \{4, 5\}$.

✓

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5

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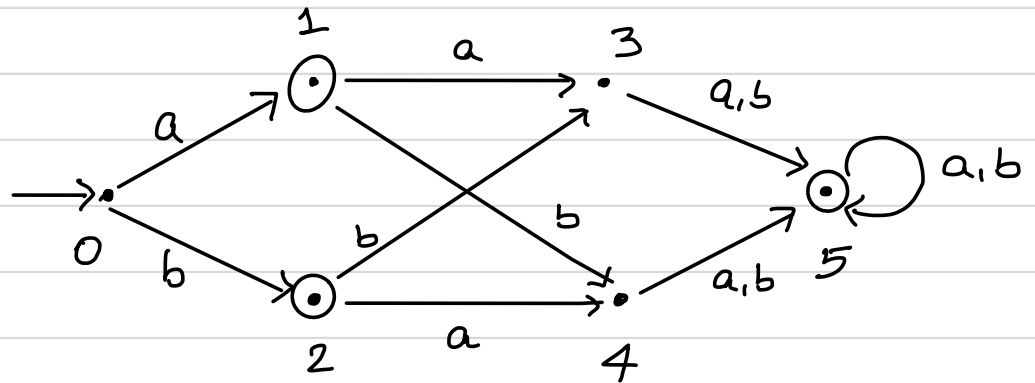
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Example.



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✓

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3

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4

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5

$\{0, 3\} \xrightarrow{a} \{1, 5\}$

$\{0, 3\} \xrightarrow{b} \{2, 5\}$

$\{1, 5\} \xrightarrow{a} \{3, 5\}$

$\{2, 5\} \xrightarrow{a} \{4, 5\}$

Theorem. The pair $\{p, q\}$ is marked by Ho Algorithm
iff $\exists x \in \Sigma^*$, s.t. $\hat{S}(p, x) \in F$ and $\hat{S}(q, x) \notin F$ or
vice versa. That is, iff $p \neq q$.

Proof. By induction.